



# Preliminary investigation update

Rejected take-off and runway overrun, Tarlton Springs - Preliminary investigation update

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INVESTIGATION WORKFLOW SUITE

## Event overview

On 18/06/2026, Saltbush Air SB512, a 50-seat twin-turboprop registered VH-SBQ, was scheduled to depart Tarlton Springs Aerodrome as the last scheduled sector of the day. On board were 44 passengers, 2 flight crew and 1 cabin crew.

### Before the flight

The aerodrome operator's only runway inspection that day was carried out at 21:35 UTC on 17/06/2026 (07:35 local on 18/06/2026), before the first scheduled arrival. It recorded the movement area as clear, and no further inspection was carried out or requested that day.

The aerodrome keeps a wildlife sighting log as a paper register in the aerodrome office. From 01/2025 to 06/2026 it contains 46 entries:

- 41 record kangaroos on or next to the movement area.
- 38 of those 41 are timed between one hour before and two hours after last light.
- 11 entries record animals on the sealed surface of runway 09/27.

The log is not reported to any operator. The aerodrome reporting officer confirmed that no airline has ever asked for it.

### The crew and the duty

The pilot in command had:

- 8,410 hours in total, 2,140 of them on type
- 3 years 1 month in command
- satisfactory results at the last cyclic simulator session (11/03/2026) and line check (02/12/2025).

The first officer had:

- 1,655 hours in total, 405 of them on type
- 14 months with the company
- a satisfactory result at the last cyclic simulator session (28/04/2026)
- two previous sectors into Tarlton Springs, both in daylight, on 09/01/2026 and 17/02/2026.

The crew signed on at 06:30 local. SB512 was the fifth sector of their duty. It had been added that morning because the rostered crew reported unfit. The duty was planned to finish at 18:35 local, a total of 12 hours 05 minutes, against a maximum of 12 hours 30 minutes for five sectors with that sign-on time. Duty and rest for both pilots in the previous 7 days were within company limits.

### Preparation and take-off clearance

The first officer reported reviewing the flight plan, charts, ERSAs and company aerodrome brief page before the flight, and reported that none of them mentioned animals. The pilot in command reported that Tarlton Springs is not on the company's list of aerodromes where a runway inspection is requested before the first departure after dark. He also reported that he had seen kangaroos on the strip there in the evening several times.

The doors closed at 07:16 UTC (17:16 local), 40 minutes behind schedule. The aerodrome was in darkness. The tower was open and rostered to close at 18:00 local.

At 07:29:10 the tower cleared SB512 for take-off on runway 09, passing the wind as 090 at 6 kt. The tower did not offer a runway inspection and the crew did not request one. No wildlife transmission appears on the tower channel in the preceding 90 minutes.

The take-off was planned for:

- a weight of 20,800 kg
- flap 10
- a dry runway
- a headwind component of 5 kt
- a calculated V1 of 112 kt.

The first officer was pilot flying.

### **The take-off roll**

Power was set at 07:30:02, with torque at 98.4% on No. 1 and 98.5% on No. 2. The pilot in command called "Power set, both engines good" at 07:30:04 and "Eighty knots" at 07:30:15.

The flight data show that:

- both engines stayed at take-off torque, within 0.1% of each other, until the power levers were retarded
- no caution, warning or engine exceedance was recorded
- the headwind component was 4 to 6 kt throughout, with no tailwind component at any point.

The pilot in command called "V one" at 07:30:29, at 112 kt.

At 07:30:31, at 116 kt, the first officer called "Roos - roos, roos, on the left, they're on the -". The occurrence notification records that she saw two kangaroos moving onto the runway from the northern side, past the 1,100 m point. She estimated the range at first sight as around 200 m, and said it could have been less. On the flight deck after the aircraft stopped she said:

> They came out of the dark on the left of the strip, crossed the centreline. I had them for - I don't know, two seconds. Maybe less.

The pilot in command stated at that time that he had not seen them before the call:

> I never had them. I was heads-in on the engine page after V one.

In his later safety report he stated that he looked up at the call and saw two large animals already on the sealed surface, moving from left to right across the centreline.

The two pilots gave differing accounts of the runway lighting:

- The pilot in command described it as a corridor of light with darkness either side.
- The first officer described it as normal. She added that, with only two daylight landings at the aerodrome to compare against, she was not well placed to judge.

### **The rejected take-off**

At 07:30:32 the pilot in command called "Stopping. I have control." The first officer acknowledged "You have control."

The reject began at 118 kt, 6 kt and 3 seconds after V1. The power levers reached flight idle within 0.6 seconds. Reconstruction from the recorded groundspeed places the aircraft 1,152 m along runway 09 at that moment, with 695 m of sealed surface remaining. In his safety report the pilot in command acknowledged that rejecting the take-off above V1 was outside procedure.

At 07:30:33 the aircraft struck both animals. The flight data recorded a lateral acceleration spike of 0.31 g to the right and a vertical spike of 1.48 g. The right main landing gear and the No. 2 propeller took the impact.

### **The stop**

Maximum braking was established at 07:30:34, 2 seconds after the decision. The operator's recorded fleet mean from decision to maximum braking is 3.2 seconds. Anti-skid was active on all four wheels.

At 07:30:36 the pilot in command called "Reverse - number two's not coming. Max brakes, max brakes." Reverse was selected on No. 1 only. No. 2 propeller RPM had been decaying since the impact, and the No. 2 reverse interlock did not make.

Between 07:30:34 and 07:30:44 the mean deceleration was  $2.6 \text{ m/s}^2$ . This is approximately 15% below the value used in the certified accelerate-stop data for that weight and runway condition, which assume reverse on both engines.

At 07:30:47 the pilot in command said "We're not stopping on it." At 07:30:51 the aircraft left the end of runway 09 at approximately 21 kt. Deceleration fell to  $1.9 \text{ m/s}^2$  in the graded gravel.

The aircraft came to rest at 07:30:57:

- 62 m into the 90 m runway end safety area
- upright and on the extended centreline
- with the nose wheel in graded gravel
- 28 m short of the end of the prepared safety area, beyond which the terrain is ungraded.

The total stopping distance from 118 kt was 757 m. Continuous tyre marks run from 1,160 m to the runway end, and the nose wheel rut continues to the stopping point.

### **After the aircraft stopped**

There were no fire indications. The pilot in command set the parking brake and stated he was not evacuating. Both engines were shut down at 07:31:26. At 07:31:34 he told the cabin to remain seated.

When the tower asked for the aircraft's situation, the pilot in command reported the overrun and advised that there were no injuries and no fire. He requested activation of the aerodrome emergency plan and a vehicle. The tower activated the plan at 07:32:10 and dispatched a vehicle.

The forward airstair was heard operating at 07:34:40, and passengers disembarked onto the ground. No evacuation was commanded and no slides were deployed. The occurrence notification records the disembarkation as 4 minutes 8 seconds after the aircraft stopped. The timeline gives the disembarkation as running from 07:34:40 to 07:38:48. Aerodrome staff attended with a vehicle, and there was no fire.

The first officer reported that during the disembarkation she saw unlit sections of the aft floor proximity strip, with the emergency lighting armed and the ceiling lights on. A functional check was carried out on 24/06/2026. [NOT ESTABLISHED: result of the floor proximity lighting functional check]

### **Consequences**

There were no serious injuries. The cabin crew member reported bruising during the disembarkation. Both kangaroos were killed; their carcasses were recovered at 1,188 m and 1,221 m from the runway 09 threshold.

The aircraft was substantially damaged:

- **No. 2 propeller:** three blades damaged beyond repairable limits, including leading-edge deformation and one blade tip separation. Animal tissue was recovered from two blade roots.

- **No. 2 engine:** subject to a shock-load inspection. The borescope inspection of both engines found no pre-existing damage and no evidence of any power anomaly before the impact.
- **Right main landing gear:** gear doors distorted. Both tyres deflated through the fuse plugs. The gear structure was within limits.
- **Brakes:** brake energy was calculated at 84% of the certified maximum. There was no brake fire or hydraulic leak, and the anti-skid system tested serviceable.
- **Lower right fuselage:** abrasion and two punctures consistent with propeller debris.

The aerodrome closed for 4 hours 20 minutes for inspection and removal of animal remains and propeller debris. The aircraft was quarantined at Tarlton Springs. The FDR and CVR were removed at 12:40 UTC on 18/06/2026. Both flight crew were stood down and tested for drugs and alcohol; both results were negative. The Head of Flying Operations suspended the evening Tarlton Springs service pending the outcome of the investigation.

## Information gathered to date

### Notification and initial actions

The operator's Duty Operations Controller notified Fleet Safety of the occurrence at 09:05 UTC on 18/06/2026. The occurrence was given a risk rating of High, and an investigation was opened the same day.

By 01:20 UTC on 19/06/2026:

- the FDR and CVR had been removed from VH-SBQ at Tarlton Springs and dispatched to the operator's flight data section
- both flight crew had been stood down and tested
- the evening Tarlton Springs service had been suspended.

### Recorded data

The investigation holds the following recorded data:

- **FDR and QAR:** readout for VH-SBQ, sector SB512.
- **CVR:** transcript for the period 07:29 to 07:34 UTC.
- **Tower:** recorded audio for channel 1 over the same period.
- **Tower occurrence log.**

The recorded data have been used to reconstruct:

- the take-off roll
- the rejected take-off
- the position of the aircraft on the runway at the moment of the decision
- the stopping sequence.

### Reports and documentation

The investigation has obtained the following documents:

- occurrence notification SBA-OCC-2026-0417
- the operator's departure record
- the flight crew and duty records for 18/06/2026
- the cabin crew member's statement
- the operator's report of the disembarkation
- the aerodrome daily inspection log for 18/06/2026

- the post-occurrence aircraft inspection and runway survey, including the borescope inspection of both engines
- a copy of the Tarlton Springs wildlife sighting log for 01/2025 to 06/2026, taken during the site visit.

The pilot in command submitted an air safety report on 19/06/2026. The report was made under the operator's confidential reporting arrangements. It has been used to understand the occurrence from his perspective and has been compared against the other sources.

The investigation has examined the operator's Part A 8.14 and the Appendix 8A list of aerodromes where a runway inspection is requested before the first departure after dark. Under the Appendix 8A listing criterion, only a reported strike at an aerodrome can lead to that aerodrome being listed. No provision asks crews to report a sighting. The investigation raised this with the operator's Safety Department on 02/07/2026. The investigation is also seeking the Safety Department's review of Appendix 8A dated 03/03/2025, to establish whether the listing criterion has been examined before.

### Site visit

Investigators visited Tarlton Springs on 23/06/2026. During the visit they:

- walked and photographed the northern aerodrome fence, finding a continuous 140 m failure and an animal pad at the 1,050 m mark
- copied the wildlife sighting log
- requested a photometric check of the runway 09/27 edge lighting.

The aerodrome reporting officer confirmed that no operator has ever requested the sighting log.

### Interviews

The investigation interviewed the pilot in command and the first officer. Both interviews were recorded and were completed by 26/06/2026. Each pilot had a nominated support person present, who did not take part. Transcripts of both interviews are held, but the interview records have not yet been finalised.

The interviews raised further matters that the investigation is examining:

- **Cabin emergency lighting:** the first officer reported unlit sections of the aft floor proximity strip during the disembarkation. A functional check of the cabin emergency lighting was carried out on 24/06/2026.
- **Wildlife information:** the first officer asked where information about kangaroos at Tarlton Springs, known to ground staff, stopped before it reached flight crews. This has been adopted as a line of enquiry.
- **Training content:** the pilot in command stated that every stop or go scenario he had been given in training involved an engine failure below V1.
- **Introduction of the evening service:** the pilot in command stated that no briefing or change that reached him accompanied the introduction of the evening service.

The cabin crew member has not been interviewed. [NOT ESTABLISHED: whether an interview with the cabin crew member is planned]

### Runway lighting

The two pilots gave differing accounts of the runway lighting. The investigation has not been able to resolve them from the evidence currently held.

- **Pilot in command:** he described a corridor of light with darkness either side. He said he could not say whether this was why he did not see the animals, and that after V1 he was looking at the engine page, as the procedure requires.
- **First officer:** she described the lighting as normal. She added that, with only two daylight landings at the aerodrome to compare against, she was not well placed to judge.

Whether the lighting affected how well the animals could be seen remains under assessment. The photometric check of the runway 09/27 edge lighting is expected to help settle this question.

### Information still awaited

The investigation is still awaiting:

- the result of the photometric check of the runway 09/27 edge lighting
- the result of the cabin floor proximity lighting functional check [NOT ESTABLISHED: result of the functional check carried out on 24/06/2026]
- the Safety Department's review of Appendix 8A dated 03/03/2025
- a response on a fleet ballast question [NOT ESTABLISHED: nature and scope of the fleet ballast enquiry]
- finalised records of the flight crew interviews.

## What has been established so far

This section sets out the events and conditions whose existence the investigation has confirmed, and the evidence for each. Where the investigation has assessed how one event or condition relates to another, that relationship is described in factual terms and with the level of confidence the investigation currently holds. None of it is a finding. The analysis continues, and any of it may change before the final report.

The section follows the sequence from the aerodrome's wildlife arrangements, through the operator's planning and controls, to the take-off roll, the rejected take-off, the strike and the stop. It then covers matters examined and ruled out, two further matters identified during the investigation that did not affect this occurrence, and one matter that could not be resolved.

## The aerodrome's wildlife arrangements

### The wildlife standard and its threshold

Tarlton Springs is a certified aerodrome. Its wildlife arrangements are governed by Aerodrome Standard AS-139 section 10.4. Subsection 10.4.1 applies to every certified aerodrome:

> 10.4.1 The operator of a certified aerodrome must take reasonable measures to reduce the risk to aircraft operations presented by wildlife on or in the vicinity of the aerodrome.

A documented plan is required only above a size threshold:

> 10.4.2 The operator of a certified aerodrome must prepare, implement and maintain a documented wildlife hazard management plan where the aerodrome is used by aeroplanes having a maximum take-off weight exceeding 27,000 kg, or serves more than 350,000 passenger movements in a calendar year.

Below that threshold, subsection 10.4.3 provides that the measures taken under 10.4.1 "need not be documented, but the aerodrome operator must be able to demonstrate what measures are in place."

Tarlton Springs meets neither limb of 10.4.2:

- its heaviest scheduled aeroplane is the 22,900 kg twin-turboprop used on the Saltbush Air service
- it handled 61,400 passenger movements in 2025.

The investigation checked this against the certified aerodrome register and the aerodrome operator's own compliance file. The aerodrome held:

- no wildlife hazard management plan
- no hazard assessment
- no dispersal programme
- no record of wildlife measures beyond the single pre-first-arrival inspection under Aerodrome Manual 6.3.

The aerodrome operator is not in breach of the standard. It is virtually certain that this arrangement existed. The standard's text and the aerodrome's traffic figures are documentary and not in dispute.

### **How the threshold relates to what followed**

The investigation has assessed whether the absence of a documented plan played a part in how the aerodrome's wildlife exposure was managed. It currently considers this likely. This relationship is held with less confidence than the others described in this section, and the evidence on it is argued rather than settled.

The case for the relationship is as follows. A documented plan is the one mechanism that would have:

- connected the aerodrome's own sighting log to the operators using the aerodrome
- put the failed northern fence on a funded programme rather than an unfunded works register.

A documented plan brings an audit trail and an owner. Several of the other controls examined in this investigation did not operate for want of those two things.

The case against is as follows:

- Subsection 10.4.1 still obliged the aerodrome operator to take reasonable measures.
- A documented plan is not the only way to repair a fence or to pass a sighting log to an airline.
- The deferrals of the fence repair were budget decisions, and there is no evidence that a documented plan would have been funded any better.

The investigation's assessment rests on the balance of these arguments.

### **The aerodrome's wildlife sighting log**

The aerodrome keeps a wildlife sighting log, described in the event overview. It is a paper register held in the aerodrome office. It is not reported to any operator, and no airline has asked to see it. Of its 41 kangaroo entries between 01/2025 and 06/2026:

- 38 are timed between one hour before and two hours after last light
- 11 record animals on the sealed surface of runway 09/27.

The aerodrome therefore held a written account of kangaroo activity on and around its runway, concentrated around last light. That account did not reach any operator using the aerodrome.

### **The northern perimeter fence**

The aerodrome operator's works register records the condition of the northern perimeter fence over three years.

- **Item WR-221, raised 04/2023:** four panels displaced and wire strained on the northern boundary, between the 700 m and 1,300 m runway markers. The entry records animals observed passing through, and recommends replacement at an estimated \$41,000.
- **Budget rounds 2023, 2024 and 2025:** the item was deferred at each round as unfunded, to be reviewed at the next cycle.
- **Item WR-263, raised 02/2026:** the same section recorded as deteriorating further.

The investigation inspected the fence on 23/06/2026. It found:

- a continuous 140 m section with the bottom two wires separated from the posts
- a clear animal pad through the fence at the 1,050 m mark.

It is virtually certain that the repair was deferred and the fence was in this condition. The deferrals are recorded in the aerodrome operator's own register, and the fence condition was confirmed on site.

The investigation considers it very likely that the condition of the fence played a part in the animals reaching the movement area. A fence in the condition found does not exclude kangaroos. The animal pad through the failed section is on the northern side of the runway, upstream of the positions where the two carcasses were recovered at 1,188 m and 1,221 m from the runway 09 threshold.

## The operator's planning for the evening service

### The route risk assessment

The operator's current route risk assessment for Tarlton Springs is RRA-041, issue 3, dated 11/2022. No issue 4 exists. It lists the hazards assessed:

> Hazards assessed: runway length and accelerate-stop margin; single runway with no lighting redundancy; no alternate within 200 nm at night; limited engineering support; heat and density altitude in summer.

Wildlife is not among them. The assessment carries this restriction:

> Operating restriction: daylight operations only pending review of night lighting and diversion fuel policy.

### The change that introduced the evening service

Change Management Record CMR-2025-118 introduced one evening rotation, Sydney to Tarlton Springs to Sydney, six days per week, from 14/07/2025. This was the first scheduled company operation into Tarlton Springs after last light. The record states:

> Assessment: schedule change only. Existing aerodrome, existing aircraft type, existing crew qualifications, existing engineering support arrangements. No new hazard identified.

>

> Route risk assessment review required: No.

>

> Approved: Head of Flying Operations, 22/05/2025.

The daylight-only restriction in RRA-041 issue 3 is not referred to anywhere in CMR-2025-118, and RRA-041 was not reopened. The Safety Department's records show no involvement in the change beyond receiving the approved record.

It is extremely likely that the change was approved in this way. Both documents are current controlled copies. The investigation has not been able to interview the approver, who has since left the company, and its confidence is qualified to that extent.

### **How the change relates to what followed**

The investigation considers it likely that approving the change without reopening RRA-041 played a part in the absence of any wildlife control for the evening service. A review of RRA-041 against a night operation would have reached, in the same pass:

- the aerodrome's wildlife exposure
- the question of whether Tarlton Springs should be subject to the operator's runway inspection requirement, described below.

That outcome is not certain. A review might have addressed only night lighting and diversion fuel, which are the two matters the restriction names.

## **The operator's runway inspection requirement**

### **What the Operations Manual requires**

The operator's Operations Manual Part A section 8.14 deals with wildlife:

> 8.14.2 At an aerodrome listed in Appendix 8A, the pilot in command shall request a runway inspection from the aerodrome operator prior to the first departure after last light, and shall not commence the take-off roll until the result of that inspection has been passed.

>

> 8.14.3 At all other aerodromes, no specific action is required. Crews should remain alert to the possibility of wildlife on the movement area.

Appendix 8A sets the criterion for listing:

> An aerodrome is listed in this appendix where a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months. The list is reviewed annually by the Safety Department.

Under 8.14.4, a strike triggers a review of the aerodrome's status against the Appendix 8A criteria.

### **Why Tarlton Springs was not listed**

As current on 18/06/2026, Appendix 8A listed nine aerodromes. Tarlton Springs was not among them and has never been listed. The list was last reviewed on 03/03/2025, about four months before the evening service began.

The only route onto the list is a reported strike at that aerodrome:

- no provision in Part A asks crews to report a sighting
- no provision admits an aerodrome to the list on any other evidence.

Tarlton Springs had had no company strike, so it could not be listed. Because it was not listed, 8.14.2 did not apply there, and no inspection was required before the evening departure. The 11 sighting log entries recording kangaroos on the sealed surface could not bring the aerodrome onto the list. They were not strikes, and they had not been reported to the company. The listing criterion therefore makes the control depend on the occurrence it exists to prevent. It is virtually certain that this arrangement existed; the manual text and the appendix listing are documentary.

### **How the requirement relates to what followed**

The investigation considers it very likely that the listing criterion played a part in the animals being on the runway during the take-off roll. Had 8.14.2 applied:

- an inspection would have been requested
- a result would have been required before the roll
- the aerodrome's staff disperse animals when they find them.

The investigation does not rate this higher because an inspection completed some minutes before departure cannot guarantee a clear runway at the time of the roll.

## No runway inspection was requested

No runway inspection was requested or carried out before SB512 departed. The evidence is as follows:

- Neither the CVR nor the tower's recorded audio contains a request for a runway inspection, an offer of one, or any mention of wildlife before departure.
- The aerodrome's daily inspection log for 18/06/2026 records one inspection, at 07:35 local before the first arrival, with the movement area recorded as clear. It records no later inspection and no request for one.

Aerodrome Manual 6.3 provides for inspections outside the scheduled time at the request of the reporting officer or a pilot in command. The aerodrome reporting officer confirmed that an inspection would have been carried out if asked.

Both crew members stated that they did not consider requesting one. The pilot in command had expected Tarlton Springs to be on the Appendix 8A list and was surprised to find it was not. No document available to either crew member prompted them to consider an inspection. The operator's procedures required none at an unlisted aerodrome, and nothing the crew had reviewed before the flight referred to wildlife at Tarlton Springs.

It is virtually certain that no inspection was requested. This rests on two independent recordings and the aerodrome's own log.

The investigation considers it likely that the absence of an inspection played a part in the animals being on the runway during the roll. An inspection would have covered the section from 1,050 m to 1,250 m, where the fence pad and both carcasses were found, and aerodrome staff disperse animals when they find them. The investigation does not rate this more highly because an inspection carried out some minutes before the roll cannot guarantee that the runway stays clear.

## Light conditions at departure

The take-off roll began at 17:30 local (07:30 UTC). Light and moon data for Tarlton Springs on 18/06/2026 (AEST, UTC+10) give:

| Event                 | Time (local)                             |
|-----------------------|------------------------------------------|
| Sunset                | 16:58                                    |
| End of civil twilight | 17:25                                    |
| Moonrise              | 22:41 (waning crescent, 14% illuminated) |

The roll therefore began 32 minutes after sunset and 5 minutes after the end of civil twilight, with no moon. The aerodrome was in full darkness. There was no natural ambient light on the strip and no town lighting within sight.

The investigation visited the site at the same time of day on 23/06/2026. It confirmed that the unlit graded strip either side of the sealed surface cannot be seen from the runway centreline. The departure fell within the period in which the aerodrome's own sighting log records 38 of its 41 kangaroo entries.

It is extremely likely that the departure took place in these conditions and within that period of peak recorded activity. The astronomical data and the sighting log are documentary. The investigation's confidence is qualified only because the sighting log is a paper register whose completeness cannot be verified.

The investigation considers it likely that the light conditions played a part in the animals being on the runway undetected until late in the roll. The animals were not seen until they were on the lit sealed surface, which follows directly from the light conditions. The timing of the departure also coincides with the peak of recorded kangaroo movement.

## The animals on the runway

Two adult kangaroos were recovered from runway 09, on its northern half:

- one at 1,188 m from the threshold
- one at 1,221 m from the threshold.

The first officer saw them from the flight deck crossing from the northern side towards the south. She estimated the range at first sight as around 200 m, and called them at 07:30:31. The investigation holds that call as recorded:

> 07:30:31 FO Roos - roos, roos, on the left, they're on the -

No runway inspection had been carried out since 07:35 local that morning.

The presence and positions of the animals are not in doubt. The investigation considers it very likely that they reached the sealed surface through the failed section of the northern fence. The animal pad at the 1,050 m mark is the most probable entry point on the physical evidence. However, the fence has failed over a continuous 140 m, and the exact crossing point is not established.

The investigation considers it extremely likely that the presence of the animals on the sealed surface led to both the rejected take-off and the strike. Each follows directly from the animals being on the runway.

## Rejected take-off procedures and training

### Procedures

The operator's Operations Manual Part B section 4.6 sets the rejected take-off criteria:

> 4.6.3 Between 80 kt and V1 the take-off shall be rejected only for: fire or fire warning; engine failure; a configuration warning; loss of directional control; or the pilot in command's judgement that the aircraft is unsafe or unable to fly.

>

> 4.6.4 At or above V1 the take-off shall be continued. The decision to continue is made before V1 is reached.

Section 4.6 contains no guidance on an obstruction seen on the runway, at any speed. It does not amplify 4.6.4 for a case in which continuing the take-off is itself hazardous.

### Training

Recurrent training syllabus TRG-SYL-07 module 4 sets this objective:

> Module 4, Rejected take-off. Objective: the crew will demonstrate a correctly executed rejected take-off for an engine failure, an engine fire and a configuration warning, each initiated below V1, and will demonstrate the decision to continue for a failure recognised at or after V1.

The module's four scripted simulator scenarios are:

- engine failure at 90 kt
- engine fire at 100 kt
- configuration warning at 85 kt
- engine failure at V1 plus 5 kt, continued.

Each is an aircraft system event. Three are rejects below V1. None presents an obstruction, an animal or any other external hazard on the runway. In the 2025 to 2026 cycle, 214 crew completed module 4. No crew has been presented with an external obstruction scenario in the simulator in the five training cycles held on file.

Both pilots stated independently that every stop or go scenario they had been given had involved an engine and had arisen below V1.

It is extremely likely that the training and procedural coverage was as described. The syllabus and manual are documentary. The investigation's confidence is qualified only because scenario records beyond five cycles were not available.

### **How the training relates to the decision**

The investigation considers it likely that the absence of training or guidance for an obstruction at or above V1 played a part in the decision to reject. Two readings of the evidence are held, and both are set out here.

The first reading is that the pilot in command had about two seconds to decide, with no briefed or practised frame for the situation he faced. The decision was the first time he had considered the question at all. On this reading, an unrehearsed decision made in two seconds is not the same decision as one made against a briefed and practised frame, and it is exactly the kind of decision the training system is intended to prepare crews for.

The second reading is that the pilot in command stated plainly that he knew the rule and departed from it deliberately. He did not reject because he was unaware of the procedure. He rejected because he judged the alternative worse. On this reading, training on obstructions would not have changed what he knew. The training gap would then be a matter affecting the fleet generally rather than something that shaped this particular decision.

The investigation's current assessment favours the first reading, on balance, while recognising the force of the second. Neither reading is offered as excusing the decision. Both are offered as ways of understanding it.

### **The decision to reject above V1**

The FDR records V1 at 112 kt at 07:30:29. Both power levers were retarded to flight idle from 07:30:32, at 118 kt. That was 3 seconds and 6 kt past V1. The CVR records the whole decision in one exchange:

- > 07:30:31 FO Roos - roos, roos, on the left, they're on the -
- > 07:30:32 PIC Stopping. I have control.
- > 07:30:32 FO You have control.

In interview and in his air safety report, the pilot in command gave his reasoning. He judged that rotating into two large animals would leave the aircraft airborne at night with damage he could not assess, over country offering no usable option. He chose the course whose outcome he could see. He acknowledged that the decision was outside procedure. It is virtually certain that the take-

off was rejected at this speed. The recorded data and audio agree, and are corroborated by his own account.

The scheduled accelerate-stop distance of 1,780 m assumes that the reject begins at or before V1. Against an ASDA of 1,847 m, this left a scheduled margin of 67 m. The investigation considers it extremely likely that beginning the stop above V1 played a part in the aircraft not stopping on the runway. Beginning the stop 6 kt and 3 seconds late uses up that margin and more, even before the loss of No. 2 reverse is taken into account.

This decision is one link in a sequence. It followed from the animals being on the runway. It was made in the absence of any training or guidance for such a situation.

## The strike

The aircraft struck both animals at 07:30:33, at 114 kt, one second after the reject began. The evidence of the strike is as follows:

- **Recorded data:** the FDR records a lateral acceleration spike of 0.31 g to the right and a vertical spike of 1.48 g.
- **No. 2 propeller:** three blades damaged beyond repairable limits, one blade tip separated, and animal tissue recovered from two blade roots.
- **Right main landing gear:** gear doors distorted.
- **Lower right fuselage:** skin abraded and punctured in two places by propeller debris.

No. 2 propeller RPM decays from the moment of impact, and the No. 2 reverse interlock did not make. It is virtually certain that the strike occurred as described. The recorded data, the physical damage and the recovered tissue all agree.

The strike took one of the aircraft's two reverse units out of use two seconds into a stop that had a scheduled margin of 67 m. It is virtually certain that this reduced the deceleration the aircraft could achieve. The measured effect is set out below.

## Deceleration and stopping distance

The certified accelerate-stop figures assume two things:

- the stop begins at or before V1
- reverse is available on both engines.

Neither held on this occasion. Reconstruction from recorded groundspeed places the aircraft 1,152 m along runway 09 when the reject began, with 695 m of sealed surface remaining. The stopping distance actually achieved from 118 kt was 757 m, which is 62 m more than the surface remaining.

Reverse was available on No. 1 only. The mean deceleration was 2.6 m/s<sup>2</sup>, approximately 15% below the value used in the certified accelerate-stop data for this weight and runway condition. The recorded data, the distance reconstruction and the tyre marks on the runway agree. It is virtually certain that the achieved deceleration was below the scheduled performance, and virtually certain that this shortfall, taken together with the late start to the stop, accounts for the overrun.

The overrun therefore has two parts:

- a stop begun 6 kt above V1
- a deceleration approximately 15% below what the scheduled figures assume.

Both were needed to carry the aircraft past the runway end. Neither alone would have done so. The investigation is treating them as a single mechanism that brought the two sequences together, rather than as separate matters.

## The crew's execution of the stop

The investigation has established that, once the decision to reject was made, the stop was carried out promptly and effectively:

- The power levers reached flight idle 0.6 seconds after the decision.
- Maximum braking was established at 07:30:34, 2 seconds after the decision. The operator's recorded fleet mean is 3.2 seconds. This response is at the fast end of the operator's own recorded distribution.
- Anti-skid was active on all four wheels throughout.
- Brake energy reached 84% of the certified maximum without a brake fire.
- Reverse was selected on No. 1 at 07:30:36. No. 2 reverse was unavailable because its propeller had been destroyed in the strike.
- The first officer made the speed callouts and the runway-end call.

The aircraft left the sealed surface at approximately 21 kt. It stopped upright and on the extended centreline, 62 m into the 90 m runway end safety area, and was disembarked without an evacuation. It is virtually certain that the stop was carried out as described; this rests on the recorded data throughout.

The investigation considers it very likely that the speed of this response reduced the severity of the outcome. The investigation estimates that a stop begun at the fleet mean response time would have carried the aircraft a further 30 to 40 m. In that case the aircraft would have been at or near the far edge of the runway end safety area, beyond which the terrain is ungraded.

## Matters examined and ruled out

### Tailwind

A tailwind above limits has been ruled out; it is very unlikely that one existed. The aerodrome observations at 06:30, 07:00 and 08:00 UTC and the FDR wind recorded during the roll all show a headwind component on runway 09 throughout. The crew's performance calculation used a 5 kt headwind, which is consistent with both sources.

### Engine power loss

A loss of power on the No. 2 engine before the reject has been ruled out; it is very unlikely that one occurred. The evidence is as follows:

- The FDR shows both engines at take-off torque, within 0.1% of each other, until the power levers were retarded at 07:30:32.
- No caution, warning or exceedance is recorded before the reject.
- The CVR contains no engine-related call.
- The borescope inspection found no pre-existing damage in either engine.
- The No. 2 propeller damage and RPM decay both begin at the moment of impact, after the reject had started.

## Other matters identified

The investigation has identified two further matters. Neither affected the overrun or its outcome. Both are described here as potential areas of risk exposure, on a preliminary basis.

### Cabin floor proximity emergency lighting

During the disembarkation, the first officer noticed sections of the aisle floor strip unlit while walking back through the cabin. A functional check was carried out on 24/06/2026, with the emergency lighting system armed and the aircraft on battery power. It found:

- two of the six floor proximity escape path segments, on the left side aft of row 16, did not illuminate
- the fault was traced to a failed ballast common to both segments.

The pre-flight emergency lighting test on this type illuminates the ceiling emergency lights and the exit signs. It does not illuminate or monitor the floor proximity segments. The defect therefore could not be detected by the crew, or by a line engineer, without a dedicated functional test. The last dedicated test was 11 months before the occurrence, within the maintenance programme's 24-month interval. It is virtually certain that the defect existed.

It is unlikely that the defect had any bearing on this occurrence:

- no evacuation was commanded
- the main cabin lights were on throughout
- the disembarkation was by the forward airstair, without urgency, and was completed without incident.

The investigation nonetheless considers it very likely that the defect is important to safety. The circumstances of this disembarkation were benign. In a night evacuation with a fire indication or smoke, two dark segments in the aft cabin could make the difference between an orderly exit and a blocked one.

The monitoring arrangement is the wider concern. A 24-month test interval, on a system the pre-flight test cannot see, means that the true state of the fleet's floor proximity lighting is not known. The same ballast is fitted across the type. Whether other aircraft in the fleet are affected has not yet been established, and the operator's engineering area is examining this.

### **Crew duty**

SB512 was the fifth sector of a duty that began at 06:30 local. That sector had been added that morning because the rostered crew reported unfit. The occurrence was 11 hours 00 minutes into the duty. The duty was planned to total 12 hours 05 minutes, against a limit of 12 hours 30 minutes for five sectors from that sign-on. The aircraft departed 40 minutes behind schedule. Both pilots described themselves in interview as tired but not struggling. It is virtually certain that the duty was as described; the roster, duty record and departure times are documentary.

The investigation has examined whether the duty affected the crew's actions, and considers it unlikely that it did. The specific ways in which fatigue would have had to act in this occurrence are each contradicted by the evidence:

- The response to the reject was at the fast end of the operator's recorded fleet distribution.
- There was no error of omission during the roll.
- Not requesting a runway inspection was not a step either crew member had been asked to consider, so it was not open to being affected by fatigue.

The investigation considers it likely that this duty pattern is important to safety in its own right. The evening Tarlton Springs rotation routinely places the last sector within the last hour of a 12-hour duty. It does so at an aerodrome with:

- no company engineering support
- no alternate within 200 nm at night
- no crewing resilience.

A single crew member reporting unfit can turn a four-sector day into a five-sector day on the morning of operation. This is a standing feature of the schedule rather than a one-day event.

## **A matter examined and not resolved**

### **Runway edge lighting intensity**

The investigation examined whether the intensity of the runway edge lighting reduced the crew's ability to see the animals. It has not been able to establish this either way. On the evidence available it is about as likely as not.

The evidence pointing towards an effect is as follows:

- The pilot in command described the lighting as brighter than he would want, forming a corridor of light with everything outside it black. He said that nothing off the sealed surface could be seen approaching.
- A photometric check on 25/06/2026 measured four edge lights on the northern side, between the 900 m and 1,200 m markers, at 118% to 131% of the nominal intensity for stage 3. The remaining lights were within tolerance.
- The four lights above tolerance are in the section of runway the animals crossed.
- The aerodrome operator's maintenance programme checks lamp serviceability but does not measure intensity.

The evidence pointing away from an effect is as follows:

- The first officer, who saw the animals, described the lighting as normal. She said she saw them clearly once they were on the sealed surface, at around 200 m. She also said that, with only two daylight landings at the aerodrome to compare against, she was not well placed to judge whether the lighting was unusual.
- The pilot in command's non-detection has an explanation that does not depend on the lighting. After V1 he was looking at the engine page, which is where the procedure directs him.

The matter could not be resolved for three reasons:

- The two pilots' accounts do not agree.
- No measurement exists of the lighting as it was on the night. The only measurement was taken seven days later.
- Nothing establishes a link between an edge light of that intensity and the detection of an object on the runway.

The measured deviation is real, and it is in the relevant section of runway. Its effect on detection has not been shown, and cannot be shown with the evidence currently available. The investigation draws no conclusion on this matter.

## **Potential areas of risk exposure**

### **Documented wildlife plan required only above a size threshold the aerodrome does not meet**

Under Aerodrome Standard AS-139, subsection 10.4.2, a documented wildlife hazard management plan is required only where an aerodrome is used by aeroplanes above 27,000 kg MTOW or serves more than 350,000 passenger movements a year. It is virtually certain that Tarlton Springs falls below both limbs. Its heaviest scheduled aeroplane is 22,900 kg, and it handled 61,400 passenger movements in 2025. The aerodrome held no plan, no hazard assessment and no dispersal

programme. Beyond the single pre-first-arrival inspection in Aerodrome Manual 6.3, it had no record of wildlife measures. The aerodrome operator is not in breach of the standard.

The investigation's current assessment is that the threshold likely influenced the occurrence. This assessment is less firm than the others in this report and is not settled. A documented plan would have provided an audit trail and an owner, and the other controls in this occurrence lacked both. It would also have linked the aerodrome's own sighting log to the operators using the aerodrome, and could have placed the failed northern fence on a funded programme rather than an unfunded works register.

There is evidence the other way. Subsection 10.4.1 still required reasonable measures, and a documented plan is not the only way to repair a fence or share a sighting log. The council's deferrals were budget decisions, and there is no evidence that a documented plan would have attracted funding.

This is a preliminary and potential area of risk exposure, subject to the completed investigation. No safety action has been recorded.

### **Perimeter fence repair deferred for three budget cycles**

A section of the northern perimeter fence at Tarlton Springs has been known to be in poor condition since April 2023, and its repair has been repeatedly deferred. The aerodrome works register item WR-221, raised in April 2023, recorded four displaced panels and strained wire between the 700 m and 1,300 m runway markers, with 'animals observed passing through'. Replacement was estimated at \$41,000. The item was deferred at the 2023, 2024 and 2025 budget rounds, and each entry reads 'defer, unfunded, review next cycle'. Item WR-263, raised in February 2026, recorded the same section deteriorating further.

The investigation's site inspection on 23 June 2026 found a continuous 140 m section with the bottom two wires separated from the posts. It also found a clear animal pad through the fence at the 1,050 m mark. The carcasses were recovered at 1,188 m and 1,221 m, on the same side of the runway.

The fence in this condition does not exclude kangaroos. On the evidence to date, the breach was very likely the route by which the two kangaroos reached the runway strip. The aerodrome's wildlife sighting log for January 2025 to June 2026 holds 41 kangaroo entries. Of these, 38 were around last light and eleven were on the sealed surface.

This is a preliminary indication of potential risk exposure and is subject to the completed investigation. While the breach remains, animals continue to have access to the movement area at the times of day when kangaroo activity has been most frequently logged.

No safety action has been recorded in response to this issue.

### **Evening service approved as a schedule change, route risk assessment not reopened**

The operator's current route risk assessment for Tarlton Springs, RRA-041 issue 3 (11/2022), restricts operations to daylight only, pending a review of night lighting and diversion fuel policy. Change management record CMR-2025-118, approved by the Head of Flying Operations on 22/05/2025, introduced a scheduled evening rotation from 14/07/2025. This was the first company operation into Tarlton Springs after last light. The change record assessed the rotation as a schedule change only and did not reference or reopen the daylight restriction. No later issue of RRA-041 exists, and the Safety Department's records show no involvement beyond receiving the approved change record.

On the evidence to date, it is extremely likely that the evening service was approved in this way. Both documents are current controlled copies. It has not been possible to interview the approver, who has since left the company.

The investigation is assessing how this relates to the occurrence. Its preliminary view is that it is likely a review of RRA-041 against a night operation would have reached the aerodrome's wildlife exposure. It is also likely that such a review would have reached the runway inspection listing in Appendix 8A of the Operations Manual. This is not certain, because a review might have addressed only lighting and fuel, which are the matters the restriction names.

This is a potential area of risk exposure and is subject to the completed investigation. The change process asked whether the change was new, not whether the existing assessment still held. As a result, it allowed an operation that directly contradicted a live restriction.

No safety action has been recorded for this issue.

### **Runway inspection required only at listed aerodromes, and the listing criterion is a strike**

The operator's Operations Manual Part A requires the pilot in command to request a runway inspection before the first departure after last light at any aerodrome listed in Appendix 8A, and not to begin the take-off roll until the result is passed. At aerodromes not on that list, the manual requires nothing. An aerodrome is listed only where a wildlife strike involving a company aircraft has been reported there in the preceding 24 months. Nine aerodromes are listed, and Tarlton Springs never has been. No provision in Part A asks crews to report a sighting, and no provision allows an aerodrome onto the list on any other evidence. The list was last reviewed on 03/03/2025, four months before the evening service began.

This is a potential area of risk exposure because the inspection requirement is the operator's own defence against this kind of occurrence, but it could not apply at Tarlton Springs. The aerodrome's paper wildlife sighting log for 01/2025 to 06/2026 contains 41 kangaroo entries, 38 of them around last light and eleven on the sealed surface. None of these could bring the aerodrome onto the list, because they were not strikes and had not been reported to the company. As a result, the control can only be activated by the type of event it exists to prevent. The same arrangement may apply at other aerodromes the operator serves after last light where animals are regularly seen but no strike has been reported.

This assessment is preliminary and subject to the completed investigation. No safety action has been recorded in response to this issue.

### **Rejected take-off training covers aircraft system failures below V1 only**

Evidence gathered to date indicates that it is extremely likely that the operator's recurrent rejected take-off training, and the corresponding guidance in Operations Manual Part B section 4.6, prepare crews only for aircraft system events. Section 4.6 lists reject items up to V1 and states that the take-off shall be continued at or above V1. The section contains no guidance on an obstruction on the runway at any speed. Both the pilot in command and the first officer independently told the investigation that the stop or go scenarios they had been given involved an engine and arose below V1.

This matters because the crew faced an external obstruction during the take-off roll, a situation that neither the manual nor the training addressed. The investigation is assessing how far this gap bore on the decision to reject the take-off 6 kt above V1, and currently considers it likely that it did.

Two readings are under consideration:

- **Training gap as an influence:** The captain had about two seconds to decide. It was the first time he had considered the question, which is the situation that practised decision-making is meant to prevent.
- **Deliberate judgement:** The captain stated that he knew the rule and departed from it deliberately because he judged the alternative to be worse. On this reading, training on obstructions would not have changed what he knew. The gap would then be a matter for the fleet as a whole rather than an explanation of this rejected take-off.

Neither reading excuses the decision. Both are relevant to understanding it. The gap potentially affects every crew completing the module.

No safety action has been recorded.

## Cabin floor proximity emergency lighting partially unserviceable

While walking back through the cabin during the disembarkation, the first officer noticed that sections of the aisle floor strip were unlit. A functional check of the cabin emergency lighting on VH-SBQ on 24/06/2026 found two of the six floor proximity escape path segments aft of row 16 on the left side dark on battery power with the system armed. The fault was traced to a failed common ballast. It is virtually certain that this condition existed.

No evacuation was commanded. The cabin lights were on throughout, and the disembarkation by the forward airstair took 4 minutes 8 seconds without incident. It is unlikely that the condition affected the overrun or its outcome.

The matter is identified as a potential area of risk exposure because it is very likely to matter in other circumstances. A fire indication, smoke or a night evacuation over the wing could turn two dark segments in the aft cabin into a blocked escape path rather than an orderly exit.

The defect could not be detected by the crew or by a line engineer without a dedicated functional test. On this type, the pre-flight emergency lighting test illuminates the ceiling lights and exit signs, but it neither illuminates nor monitors the floor segments. The last functional test was 11 months before the occurrence, within the 24-month programme interval. The same ballast is fitted across the type, so the condition of the floor proximity lighting on other aircraft is not known.

This assessment is preliminary and subject to the completed investigation. No safety action has been recorded for this issue.

## Open lines of enquiry

The investigation is continuing. This section describes the lines of enquiry still being pursued and the evidence and analysis still needed. It then deals separately with one matter that has already been examined and could not be settled. None of this material is a finding, and the direction of an enquiry does not show what it will conclude.

## Assessment still to be completed

The investigation has not yet completed its assessment of how each event and condition relates to the others. The relationships described earlier in this report are preliminary. Some of them, including the influence of the aerodrome standard's size threshold and of the rejected take-off training, were argued with evidence on each side and may be revised as further evidence is obtained.

## Flow of wildlife information from the aerodrome to flight crews

During interview, the first officer asked where information about kangaroos at Tarlton Springs, known to ground staff, stopped before it reached flight crews. The investigation has adopted this question as a line of enquiry. It covers:

- the aerodrome's wildlife sighting log, which is held as a paper register and has never been requested by an operator
- the absence of any provision in the operator's Part A asking crews to report a sighting
- the operator's aerodrome brief material, which the first officer reported made no mention of animals.

The completeness of the sighting log cannot be verified from the paper register alone. The investigation is therefore treating the 41 kangaroo entries as a minimum picture of wildlife activity rather than a complete one.

## The operator's review of Appendix 8A

The investigation raised the Appendix 8A listing criterion with the operator's Safety Department on 02/07/2026. It is seeking the Safety Department's review of Appendix 8A dated 03/03/2025, to establish whether the listing criterion has been examined before. That review took place four months before the evening service began.

## Approval of the evening service

The investigation has not interviewed the Head of Flying Operations who approved CMR-2025-118, because that person has since left the company. The documentary evidence for how the change was approved is held as current controlled copies. Without an account from the approver, the investigation cannot establish what was considered when the change was assessed as a schedule change only. [NOT ESTABLISHED: whether an interview with the former approver will be sought]

## Rejected take-off training history

The investigation has examined the rejected take-off scenarios in the five training cycles for which information is available. Scenario information for earlier cycles is not available. [NOT ESTABLISHED: whether earlier scenario information is being sought]

## Route taken by the animals

The northern fence has failed over a continuous 140 m section. The animal pad at the 1,050 m mark is the most probable entry point on the physical evidence, but the exact point at which the two kangaroos crossed the fence has not been established.

## Floor proximity lighting across the fleet

The failed ballast found on VH-SBQ is fitted across the aircraft type. The floor proximity segments are not illuminated or monitored by the pre-flight emergency lighting test, and the functional test interval is 24 months. As a result, the condition of this lighting on the operator's other aircraft of the type is not known. The investigation has referred the fleet-wide question to the operator's engineering function and is awaiting its response.

## Interviews and statements

Transcripts of the interviews with the pilot in command and the first officer are held but have not yet been finalised. The cabin crew member has provided a statement but has not been interviewed. [NOT ESTABLISHED: whether an interview with the cabin crew member is planned]

## **A matter examined and not resolved: runway edge lighting and detection of the animals**

The investigation has examined whether the intensity of the runway edge lighting reduced the crew's ability to detect the animals. It has not been able to settle the question. On the evidence available, it is about as likely as not that the lighting condition proposed existed in a form that affected detection. The matter is therefore not a finding in either direction.

### **Evidence that supports the proposition**

- The pilot in command described the lighting as a corridor of light with darkness either side, and brighter than he would want. He said nothing off the seal could be seen coming.
- A photometric check on 25/06/2026 measured four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of nominal for stage 3. This is the section of runway the animals crossed.
- The aerodrome's maintenance programme checks lamp serviceability but does not measure intensity.

### **Evidence against the proposition**

- The first officer saw the animals. She described the lighting as normal and said she saw them clearly once they were on the sealed surface, at around 200 m.
- The first officer also said that, with only two daylight landings at the aerodrome to compare against, she was not well placed to judge whether the lighting was unusual.
- After V1, the pilot in command was looking at the engine page, where the procedure places him. That explains why he did not see the animals without any reference to the lighting.

### **Why the matter remains open**

The two pilots' accounts do not agree. The photometric check was carried out seven days after the occurrence, so there is no measurement of what the lights were doing on the night. The check shows a deviation from nominal intensity in the relevant section. Nothing in the evidence links an edge light at 131% of nominal to how well an object on the runway can be detected.

The investigation cannot close the question with the evidence currently available. [NOT ESTABLISHED: what further evidence or analysis, if any, could resolve the effect of edge lighting intensity on detection]

## **Investigation status and next steps**

### **Where the investigation stands**

The investigation was opened on 18/06/2026, the day of the occurrence, with a risk rating of High. That rating has been reviewed and is still considered suitable. Terms of reference have been issued, and a timeframe has been set that includes acquittal of findings. [NOT ESTABLISHED: target dates for completion of the analysis and issue of the final report]

Evidence gathering is substantially complete. The investigation holds:

- the recorded flight, cockpit and tower data

- weather records
- photographs of the site and damage
- the operator's manuals and procedures
- personnel, training, duty and maintenance records
- change management records
- witness and occupant statements
- third-party evidence
- the results of the site visit and component examination.

A fatigue analysis and drug and alcohol testing were required. Both flight crew returned negative drug and alcohol results.

Some containment measures are already in place. On 18/06/2026 the Head of Flying Operations suspended the evening Tarlton Springs service pending the outcome of the investigation. [NOT ESTABLISHED: current status of the evening service suspension] Matters outside the scope of the investigation have been identified and referred. The listing criterion for Appendix 8A was raised with the operator's Safety Department on 02/07/2026.

### **Analysis**

The investigation has mapped the hypotheses and argued the contested relationships with evidence on each side. It has documented the hypotheses that were ruled out, with the reasons, and has recorded the positive events.

The assessment is not yet complete for every event and condition. Two matters are still unsettled:

- the effect of runway edge lighting intensity on detection of the animals
- the fleet-wide condition of the floor proximity emergency lighting.

The relationships described in this update are preliminary until that assessment is finished.

### **Immediate next steps**

The investigation's next steps are to:

- obtain the Safety Department's review of Appendix 8A dated 03/03/2025, and establish whether the listing criterion has been examined before
- obtain the operator's engineering response on the condition of floor proximity lighting across the fleet, given that the ballast type that failed on VH-SBQ is fitted across the aircraft type
- consider what further evidence or analysis, if any, could resolve the question of edge lighting intensity and detection. [NOT ESTABLISHED: what further evidence or analysis is proposed]
- finalise the records of the interviews with the pilot in command and the first officer
- decide whether to interview the cabin crew member. [NOT ESTABLISHED: whether an interview with the cabin crew member is planned]
- decide whether to seek an account from the former Head of Flying Operations who approved CMR-2025-118. [NOT ESTABLISHED: whether an interview with the former approver will be sought]
- complete the assessment of every event and condition, with a confidence recorded for each
- once the analysis is complete, carry out peer review by the investigation team and a stakeholder review, and then draft the final report. Findings will then be assigned to owners with timeframes.

**Decision supported by this update**

This update supports a decision on whether the investigation should continue to a fuller report, or be closed because it is evident that no systemic or organisational factors are involved and the potential for further learnings is limited.

The potential areas of risk exposure identified so far concern organisational and regulatory arrangements. They include:

- the aerodrome standard's size threshold for a wildlife plan
- the deferred repair of the perimeter fence
- how the evening service was approved
- the Appendix 8A listing criterion
- the scope of rejected take-off training.

Several lines of enquiry into these arrangements are still open. On the information available, the evidence does not indicate that systemic or organisational factors can be ruled out, or that the potential for further learnings is limited.

The decision whether to continue to a fuller report or close the investigation rests with the responsible safety manager.

**Distribution**

| <b>Title</b>                                 | <b>Name</b> |
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